

CareNest

AI Field Assistant for Community Health Workers

Hackathon MVP Product Requirements Document

One-line definition

CareNest helps community health workers in low-resource settings capture patient visits by voice, use AI to turn natural notes into structured records, preserve patient history across visits, and carry relevant information forward for follow-up or referral.

Product name	CareNest
Version	2.0 - Hackathon MVP
Date	19 September 2026
Track	Social Impact / Non-Profit
Primary user	Community Health Worker (CHW)
Secondary users	Supervising health worker; receiving health worker
Organizational home	Community health facility / PHC / NGO programme
Primary setting	Rural and low-resource communities in Nigeria
Product form	Mobile-first Progressive Web App (PWA), installable on Android and desktop

1. Executive Summary

CareNest is a mobile-first, offline-aware field assistant for Community Health Workers (CHWs) working in rural and low-resource communities. It reduces the burden of turning real-world patient conversations and observations into structured documentation.

The product lets a CHW speak naturally or type a short note, uses AI to transcribe and structure the information, requires the CHW to review the AI draft, and then saves the confirmed encounter to the patient record. When the patient returns, another authorized health worker can retrieve the patient history instead of starting from zero. When a referral is required, CareNest creates a concise handoff summary and a Care Pass / referral code that the receiving facility can use to retrieve the relevant information.

Product boundary

CareNest is not a diagnostic or prescribing system. The AI organizes information and prepares documentation; the CHW or other qualified health worker remains responsible for clinical decisions and final confirmation.

2. Hackathon Strategy

The hackathon rewards a specific user, a specific painful problem, meaningful AI, a working end-to-end prototype, and a believable next step. CareNest is deliberately scoped around one workflow rather than a broad healthcare platform.

Specific user	Community Health Worker
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Specific pain	Fast, reliable documentation and continuity of patient information across visits and referrals
AI unlock	Natural voice/text input is converted into structured, reviewable documentation and referral summaries
Core demo	One patient journey: visit → AI documentation → saved history → return visit → referral → receiving facility
Low-resource design	Mobile-first, installable PWA, voice-first input, offline-aware storage and simple UI
Future path	Pilot with one community facility/NGO programme, measure time saved and referral continuity, then expand

3. Problem Statement

Community Health Workers are often the first point of care for people in rural and underserved communities. Their work produces valuable patient information, but capturing that information can be slow and fragmented.

- A visit often begins as natural conversation, while organizations need structured records.
- Typing lengthy notes on a phone can be burdensome for a field worker moving between patients or homes.
- Poor or intermittent internet connectivity can interrupt digital workflows.
- When a patient returns, the next health worker may need to reconstruct the story from memory, a paper file, or incomplete information.
- When a patient is referred, relevant context may not move cleanly with the patient to the next point of care.

Core problem: CHWs need a simple way to capture accurate encounter information without excessive form-filling, while keeping that information available for continuity, follow-up, and referral handoff.

4. Product Opportunity

CareNest sits between the CHW's natural workflow and the organization's need for structured records.

The transformation

Natural communication → AI-assisted structure → Human review → Persistent patient history → Actionable follow-up / referral

5. Product Vision

Make field documentation as natural as talking, without sacrificing structure, accountability, privacy, or continuity of care.

6. Product Principles

Principle	Meaning
Field-first	Design for a phone, fast interactions, one-handed use where practical, and low-bandwidth environments.
Human-in-the-loop	AI creates a draft; the health worker reviews, edits, and confirms the final record.
Structure without friction	Let workers communicate naturally before asking them to fit the information into fields.
Offline resilience	Connectivity problems should affect synchronization, not the ability to capture a visit.

Principle	Meaning
Continuity over duplication	One patient history should be retrievable across authorized encounters at the organization.
Action over documentation	A completed encounter can produce a follow-up or a referral handoff when needed.
Source transparency	Clearly distinguish patient-reported information, CHW observations, and AI-generated structure or suggestions.
Narrow before broad	Win the hackathon with one exceptional workflow rather than a collection of half-built healthcare features.

7. Target Users & Roles

Role	What they do	What they need
Primary: Community Health Worker	Captures visits, reviews AI drafts, views patient history, creates follow-ups and referrals.	Must be fast, simple, voice-friendly, and usable with weak connectivity.
Secondary: Receiving Health Worker	Retrieves a referral or Care Pass and reviews the information intentionally shared by the referring facility.	Needs a fast handoff view; should not need to search through unrelated records.
Secondary: Supervisor	Monitors visits, pending follow-ups, handoffs, and sync status at a high level.	Useful but not the hero workflow for the hackathon demo.
Customer / Organization	Community health facility, PHC network, NGO or programme deploying CHWs.	Provides the organizational workspace and record ownership context.

Important product decision: CareNest is facility/organization-based, but CHWs and nurses are the people who actually use the product. The patient record belongs to the organization's workspace, not to one individual worker account.

8. Core User Workflow

End-to-end MVP flow

Facility / Workspace → CHW Login → Search or Create Patient → Record Visit → Voice/Text → AI Structure → Review/Edit → Confirm → Save → Patient Timeline → Follow-up OR Refer → Care Pass → Receiving Facility Retrieval

8.1 Visit workflow

1. CHW logs into the facility workspace.
2. CHW searches for the patient. If the patient is new, the CHW creates a basic patient profile.
3. CHW starts a new encounter and chooses Voice or Text.
4. CHW speaks naturally or types a short note describing the visit, observations, and relevant information.
5. AI transcribes and structures the input into a reviewable encounter draft.
6. CHW checks the draft, edits any mistakes, and explicitly confirms it.
7. CareNest saves the confirmed encounter to the patient timeline.
8. CHW chooses a next action: no action, follow-up, or referral/handoff.

8.2 Return visit workflow

9. Patient returns to the facility or another authorized worker sees the patient.
10. Worker searches the patient record.

11. Previous encounters appear as a chronological timeline.
12. Worker opens the relevant encounter(s), gets context, and starts the new visit without recreating the entire history.

8.3 Referral / care handoff workflow

13. CHW selects Refer Patient after making the clinical decision that higher-level care is needed.
14. CareNest prepares a structured referral summary using the confirmed record and only the relevant information.
15. CHW reviews and confirms the referral summary.
16. CareNest generates a Care Pass / referral ID, optionally represented as a QR code.
17. Receiving health worker scans or enters the code.
18. The receiving worker sees the authorized referral information and can continue the handoff.

Referral guardrail

AI prepares the handoff documentation; it does not decide whether a patient should be referred. The decision remains with the health worker.

8.4 Offline workflow

19. When internet is available, normal data submission occurs.
20. When connectivity is lost, the app keeps the active workflow available and stores eligible visit data locally on the device.
21. The CHW sees a clear "Saved offline / Pending sync" state.
22. When connectivity returns, the app syncs the pending record to the backend and updates the status to "Synced".

Hackathon implementation target: demonstrate the user experience of offline capture and pending sync.

Production-grade conflict resolution and multi-device synchronization are future hardening tasks.

9. MVP Feature Requirements

ID	Feature	Requirement	Priority
FR-01	Authentication & facility workspace	CHW can sign in and operate inside the correct organization/workspace.	Must
FR-02	Patient registration	Create a basic patient profile with unique patient ID.	Must
FR-03	Patient search	Find an existing patient quickly by name, ID or phone where appropriate.	Must
FR-04	Patient timeline	Display previous confirmed encounters chronologically.	Must
FR-05	Visit capture	Capture free-form text and/or voice notes for a new encounter.	Must
FR-06	AI structuring	Convert natural-language input into a structured encounter draft.	Must
FR-07	Human review	CHW can edit AI output and must confirm before final save.	Must
FR-08	Source attribution	Show what is patient-reported, observed by CHW, or generated as AI structure/suggestion when applicable.	Should
FR-09	Follow-up task	Create a simple follow-up action with a date/status.	Should
FR-10	Referral summary	Generate a concise handoff summary from the confirmed encounter.	Must
FR-11	Care Pass	Generate a referral code / QR that identifies the handoff	Must

ID	Feature	Requirement	Priority
		request.	
FR-12	Receiving facility view	Retrieve authorized referral information using the Care Pass.	Must
FR-13	Offline queue	Show locally saved items and sync status when offline.	Must
FR-14	PWA installation	Allow the app to be installed from the browser as a standalone app with an icon.	Must
FR-15	Supervisor dashboard	High-level counts for visits, follow-ups, referrals and sync status.	Should / Secondary

10. AI Requirements & Safety Guardrails

10.1 Where AI is used

- Speech-to-text: convert a CHW voice note into text.
- Information structuring: map the note into defined fields such as complaint, duration, observations and follow-up needs.
- Encounter summarization: create a concise reviewable summary of a confirmed encounter.
- Referral summarization: draft the relevant handoff information from confirmed records.

10.2 AI must not

- Diagnose a patient.
- Prescribe medication or recommend a treatment plan as a clinical authority.
- Invent symptoms, vitals, medications, history or other facts that were not provided.
- Silently overwrite CHW-entered information.
- Automatically decide that a patient needs referral.

10.3 Human-in-the-loop rule

Every AI-generated encounter or referral draft must display a clear review state. The CHW must be able to edit the information and explicitly confirm the final version before it becomes part of the official record.

10.4 Structured output

AI output should be validated against a defined schema before saving. Example fields: chief concern, duration, symptoms/concerns, vitals, observations, patient-reported information, CHW observations, follow-up actions, referral reason, and source attribution.

11. PWA & Installability Requirements

CareNest should be delivered as a Progressive Web App (PWA), so a CHW does not need to type a website address every time. After visiting CareNest once in a supported browser, the worker can install it to the device home screen and launch it like an app.

Requirement	Expected behavior
Installable	Provide a web app manifest with app name, icons, theme/background metadata and standalone display mode.
Service worker	Cache the app shell and critical assets so the application opens reliably after installation and can support offline-aware workflows.
App icon	Use a clear CareNest icon on the Android home screen / desktop app launcher.

Requirement	Expected behavior
Mobile-first	Design for small Android screens, touch targets, simple navigation and low cognitive load.
Offline storage	Use IndexedDB or equivalent local storage for eligible draft/pending records.
Connectivity status	Show Online, Offline, Pending Sync and Synced states clearly.
Fast launch	Open directly to CHW Home or the last safe authenticated state where technically appropriate.

PWA terminology

The correct term is PWA: Progressive Web App. It gives the team the app-like installation experience without requiring a native Android build for the hackathon.

12. Core Screens

Screen	Purpose
1. Login / Workspace	Facility identity + CHW account sign-in.
2. CHW Home	New Patient, Find Patient, pending sync, follow-ups and minimal daily counts.
3. Patient Search	Search and select existing patients; create new patient when needed.
4. Patient Profile	Basic demographics + encounter timeline + open new encounter.
5. New Visit	Voice / Text capture plus optional quick fields such as vitals.
6. AI Review	Show transcript/structured draft, source labels, edit controls and Confirm.
7. Encounter Detail	Confirmed record with timestamp, worker, notes and next action.
8. Referral	Review referral reason, generated summary and destination.
9. Care Pass	Referral ID and QR code for handoff.
10. Receiving Facility	Enter/scan Care Pass and view the authorized referral summary.
11. Supervisor Dashboard	Simple operational view; secondary for demo.

13. Data Model (Hackathon Simplification)

The system should be organized around the facility, with health workers acting inside the facility workspace.

Entity	Core fields / purpose
Facility	id, name, location, type
User / Health Worker	id, facility_id, name, role, authentication info
Patient	id, facility_id, name, age/date of birth, sex, phone where appropriate
Encounter	id, patient_id, facility_id, worker_id, date/time, structured fields, source metadata, confirmation status
Follow-up	id, encounter_id/patient_id, action, due date, status
Referral	id, encounter_id, source facility, destination facility, reason, handoff summary, status, code/QR token
Sync Queue	local item id, entity type, sync status, timestamps, retry metadata

14. Technical Architecture

The stack should optimize for speed of implementation, reliable AI integration, and a deployable demo rather than production-scale complexity.

Layer	Hackathon choice
Frontend	React / Next.js + Tailwind CSS

Layer	Hackathon choice
Installability	PWA manifest + service worker + responsive mobile-first UI
Local/offline storage	IndexedDB for drafts and pending sync queue
Backend	Node.js with Express or Fastify
Database	PostgreSQL + Prisma
Validation	Zod for request and AI output schemas
Speech-to-text	Whisper or Groq Whisper
LLM / AI structuring	Groq-hosted LLM with strict JSON/schema output
Deployment	Vercel for frontend; Render/Railway or equivalent for backend

15. Privacy, Security & Data Handling

- Use role-based access so users only see records appropriate to their role and facility context.
- Transmit data over HTTPS in deployment environments that support it.
- Avoid exposing sensitive patient information in URLs, logs, screenshots, or demo materials.
- Use demo patient data for the hackathon rather than real patient data.
- Treat the Care Pass as an access token, not as a QR code containing a complete medical record.
- For production, add consent, retention, audit logs, stronger authentication, encryption strategy, access governance and compliance review before handling real patient data.

16. What Is Explicitly Out of Scope for the Hackathon

- Full EHR / hospital information system.
- Clinical diagnosis or autonomous treatment recommendations.
- Medication prescribing or pharmacy management.
- Billing, insurance or payments.
- Telemedicine platform.
- Ambulance dispatch or transport marketplace.
- Complex hospital appointment scheduling.
- Patient-facing mobile app / patient portal.
- Large multi-facility health exchange with complex interoperability standards.
- Production-grade conflict resolution across many devices.

These can be future roadmap items only after the core workflow proves useful.

17. Success Metrics for the Prototype

Metric	Prototype target
Core success metric	A complete patient journey can be executed end-to-end in the prototype.
Documentation speed	Demonstrate a reduction in steps/time versus manually filling a long form.
AI usefulness	AI structures the CHW's input accurately enough that the worker only needs to review/edit rather than rewrite.
Continuity	A return visit retrieves the earlier confirmed encounter within seconds.
Referral handoff	A receiving worker can retrieve a concise, relevant handoff using the Care Pass.
Offline experience	A visit can be captured while offline and visibly enter a pending-sync state.
Installability	The app can be installed to a device and relaunched from the home screen without retyping the web address.

18. Hackathon Demo Story

The demo should follow one patient rather than showing disconnected screens. The point is to make the judge see the entire value chain in one continuous story.

23. Introduce Maria, a patient in a rural community who visits a CHW.
24. CHW creates or finds Maria's patient record.
25. CHW taps the microphone and describes the visit naturally.
26. AI transcribes and structures the encounter.
27. CHW reviews, corrects, and confirms the record.
28. Maria returns later; the CHW retrieves her previous encounter instead of starting again.
29. CHW chooses to refer Maria for higher-level care.
30. CareNest prepares a referral summary and generates a Care Pass.
31. A receiving health worker enters/scans the Care Pass and retrieves the authorized handoff information.
32. Close with the core value: the CHW spends less time fighting forms, and the patient's story can move forward with the care.

Demo rule

Do not spend the majority of the demo on dashboards. The hero is the patient journey: Voice → AI → Review → Save → Return → Retrieve → Refer → Receive.

19. Why This Can Stand Out

- Clear user: Community Health Worker, rather than “everyone in healthcare”.
- Clear setting: rural / low-resource community care in Nigeria.
- AI has a concrete job: speech-to-text, structuring and handoff summarization.
- The facility-based record model supports continuity when different workers see the same patient.
- The PWA format makes deployment realistic on ordinary Android phones without requiring an app-store install.
- Referral is treated as a continuation of the patient story, not as a separate administrative form.
- The prototype remains narrow enough to be working and demoable within the hackathon time constraint.

20. Post-Hackathon Roadmap

Phase	Next step
Phase 1 - Pilot	Test with one community health facility or NGO programme; observe real CHW workflows and refine the UX.
Phase 2 - Reliability	Strengthen offline sync, audit logs, role permissions, device recovery and data governance.
Phase 3 - Referral network	Enable approved partner facilities to receive referral handoffs with stronger access control and interoperability.
Phase 4 - Scale	Expand across additional facilities and programmes; add reporting and integration capabilities based on validated demand.

21. Final Product Definition

CareNest

An installable, offline-aware PWA for Community Health Workers that turns natural patient-visit conversations into AI-assisted structured records, keeps confirmed history available across visits, and carries relevant information forward for follow-up and referral.

22. Hackathon Build Acceptance Checklist

- ☐ The CHW can sign in and reach a patient quickly.
- ☐ The CHW can create or find a patient.
- ☐ The CHW can capture a visit by voice and/or text.
- ☐ AI visibly transforms the input into structured information.
- ☐ The CHW can review/edit and confirm before saving.
- ☐ The patient timeline shows the confirmed encounter.
- ☐ A second visit can retrieve previous history.
- ☐ A referral summary can be generated from the confirmed record.
- ☐ A Care Pass / referral code can be generated and retrieved at the receiving side.
- ☐ The app demonstrates a clear offline/pending-sync state.
- ☐ The app can be installed as a PWA with a home-screen icon.
- ☐ The entire demo can be completed smoothly without switching to unrelated features.